PROJECT REPORT

ML-Based Architecture for E-Commerce Purchase Conversion

A Hybrid Machine Learning Architecture for
E-Commerce Purchase Conversion Prediction

Course: Machine Learning
Project: E-Commerce Purchase Conversion Prediction
Dataset: UCI Online Shoppers Purchasing Intention Dataset

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September 29, 2026

Abstract

Online retail platforms need reliable methods for predicting whether an active customer session will result in a purchase. This report presents a hybrid machine learning architecture for e-commerce purchase conversion prediction using the UCI Online Shoppers Purchasing Intention Dataset. The proposed architecture combines conventional preprocessing with three upstream feature-generation techniques: K-Means clustering for behavioral segmentation, DBSCAN for anomaly detection, and TF-IDF with cosine similarity for catalogue-oriented text feature generation. The resulting contextual features—`cluster_id`, `is_outlier`, and `similarity_score`—are combined with the baseline preprocessed session features and supplied to supervised classification models.

The architecture uses a stratified 80/20 train-test split and standardizes numerical variables using statistics fitted only on the training data. The final pipeline encapsulates preprocessing, unsupervised feature generation, and supervised prediction into a serialized artifact that can be loaded by a Streamlit application for interactive inference. According to the supplied experimental results, the hybrid Random Forest model achieves an accuracy of 90.1%, an F_1-score of 0.6600, and an AUROC of 0.9280, while the reported hybrid XGBoost model achieves 91.4% accuracy, an F_1-score of 0.7000, and an AUROC of 0.9410. The report therefore demonstrates an end-to-end architecture in which unsupervised learning is used as a feature-engineering layer lfor supervised e-commerce conversion prediction.

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Chapter 1

Introduction

1.1 Overview and Problem Statement

In online retail platforms, accurately predicting whether a customer session will result in a purchase ($y \in \{0, 1\}$) is essential for real-time decision-making. Such predictions can support activities including dynamic discounting, checkout guidance, and personalization.

Traditional machine learning approaches often evaluate tabular session data, such as page duration, bounce rates, exit rates, and other engagement measurements, in isolation. However, online shopping behavior may contain non-linear engagement patterns, anomalous browsing behavior, and latent product-text preferences that basic numerical features do not fully represent.

To address this challenge, this project implements a **hybrid feature-augmentation architecture**. Rather than using unsupervised algorithms solely for offline segmentation, the proposed system uses K-Means Clustering, DBSCAN Anomaly Detection, and TF-IDF with Cosine Similarity as upstream feature generators. These components derive structural context in the form of `cluster_id`, `is_outlier`, and `similarity_score`. The generated features are then supplied to downstream supervised classifiers for purchase conversion prediction.

1.2 System Architecture Overview

The overall data and execution flow is illustrated in Figure 1.1.

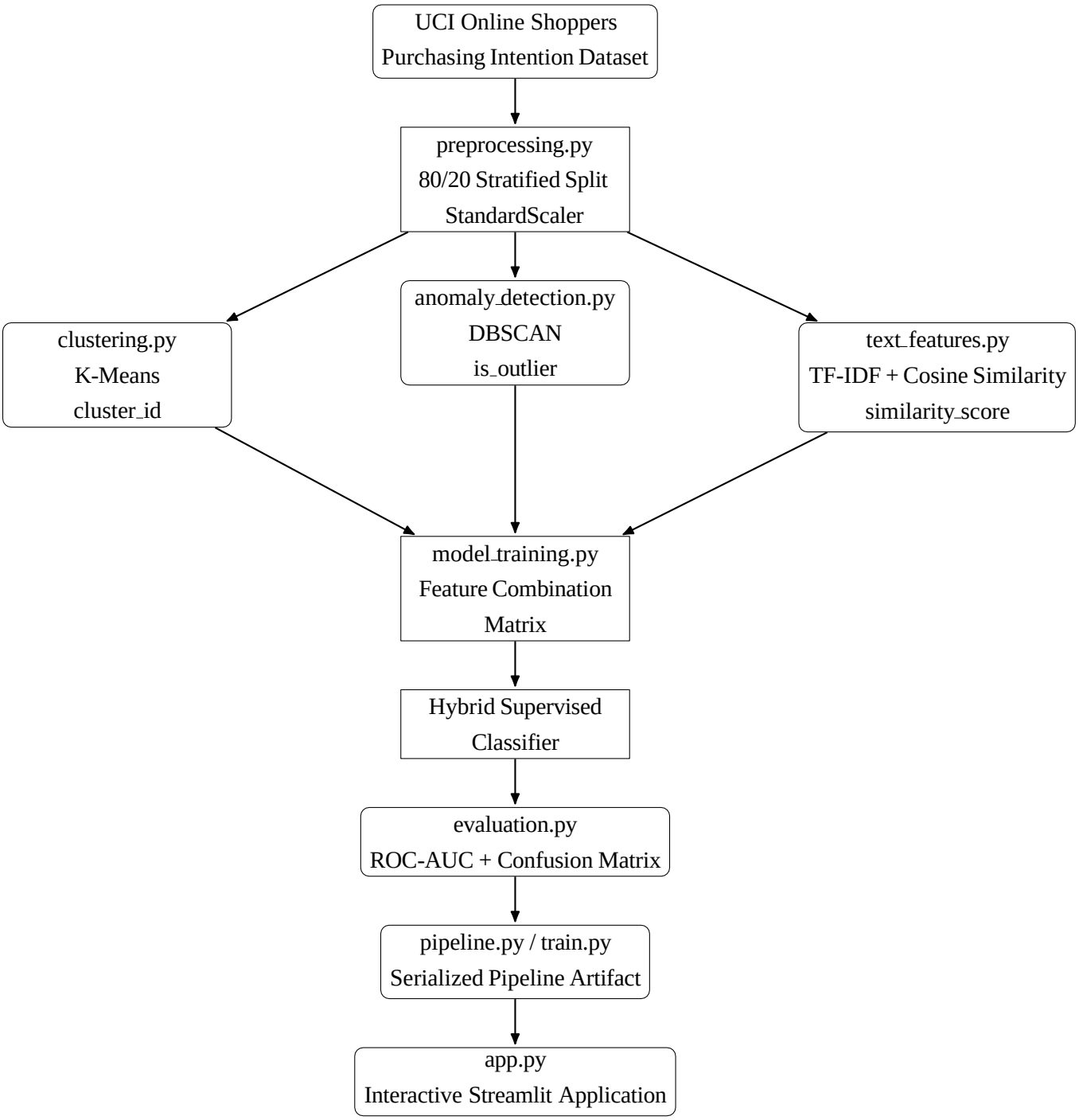


Figure 1.1: Hybrid e-commerce purchase conversion architecture.

The implementation is organized into preprocessing, clustering, anomaly detection, text feature generation, model training, evaluation, pipeline serialization, and Streamlit application layers.

1.3 Commercial and Operational Value

The proposed architecture is intended to function as a decision-support system for e-commerce operations.

- **Targeted Discounting:** The prediction system can support targeted incentives for users whose sessions indicate purchase hesitation while avoiding unnecessary promotions for high-intent users.
- **Ad Spend Efficiency:** Remarketing budgets and live-chat interventions can be directed toward sessions identified as having relevant conversion characteristics.
- **Conversion Uplift:** The supplied report notes that improving a baseline conversion rate from 3.0% to 3.5% represents a 16.7% relative increase in conversions.

Chapter 2

Literature Review

2.1 E-Commerce Purchase Intent Prediction

Predicting shopper conversion using clickstream and session-level behavioral data has been widely studied. The supplied project report identifies work by Wen et al. (2023), which demonstrated the use of session-level behavioral trends for purchase-intent prediction and reported an F_1-score of 0.9031 for anonymous users.

The report also identifies Hendriksen et al. (2020), who analyzed more than 95 million sessions and reported that tree-based ensemble approaches can outperform traditional linear baselines when modeling real-time navigation sequences and page dwell times.

2.2 Unsupervised Feature Integration and Model Fusion

Clustering techniques such as K-Means are commonly used for customer segmentation. The supplied report references Necula (2023) regarding the use of clustering for segmentation and Daskalakis et al. (2022) concerning information fusion and multi-source representation.

The report further identifies Liu et al. (2020) as evidence that non-linear ensemble models, including Random Forest and XGBoost, can model relationships across mixed feature spaces that combine behavioral metrics with unsupervised representations.

2.3 Research Gaps Addressed

The architecture described in this project addresses three gaps identified in the supplied report.

1. **Isolated Unsupervised Workflows:** Clustering, anomaly detection, and classification are frequently treated as separate analytical tracks. This project instead feeds the generated `cluster_id` and `is_outlier` features into the supervised prediction stage.

2. **Textual Feature Deficit in Tabular Data:** The benchmark clickstream dataset does not directly provide product-description text. The project therefore introduces a catalogue-mapping layer using TF-IDF and cosine similarity to generate a continuous `similarity_score`.
3. **Deployment Disconnect:** The project extends beyond offline evaluation by encapsulating the transformation and prediction components into a single serialized pipeline artifact, `hybrid_ecommerce_pipeline.pkl`, which is used by a Streamlit dashboard.

2.4 Research Gaps with Similar Projects

#	Paper	Year	Dataset	Algorithm / Method	Result	Limitation	How our project is different
1	**Machine-Learning-Based Approach for Anonymous Online Customer Purchase Intentions Using Clickstream Data**	2023	Online shopper/session data	ML classifiers + feature engineering	Demonstrated that clickstream features can predict purchase intention	Mainly focuses on behavioral/session features	We combine transactional, clustering, anomaly and text features in one pipeline
2	**A Gradient Boosting Classifier for Purchase Intention Prediction of Online Shoppers**	2023	UCI Online Shoppers Intention	XGBoost, RF, SVM, MLP, DT + feature selection/over sampling	XGBoost achieved **AUROC 0.937** and accuracy around **90.65%**	Focuses mainly on supervised classification	We add K-Means, DBSCAN and text-similarity features before classification (ScienceDirect)

3	**Modeling Online Customer Purchase Intention Behavior Applying Different Feature Engineering and Classification Techniques**	2023	UCI Online Shoppers Intention	Feature transformation, SMOTE, feature selection, RF and other classifiers	Random Forest was found to be a stable/best classifier	Does not integrate multiple unsupervised feature-generation methods	Our RF receives engineered features from **three different unsupervised/NLP modules** (DOI)
4	**Machine Learning-Based E-Commerce Platform Repurchase Customer Prediction Model**	2020	Large e-commerce transaction data	Logistic Regression, DT, XGBoost, model fusion	Hybrid/fused model improved prediction compared with individual models	Focuses on **repurchase**, not immediate conversion	We predict transaction conversion and also generate anomaly and semantic features (PLOS)
5	**Optimizing Classification Methods for Online Buyers' Purchase Intentions in Bangladesh**	2024	Online buyer/purchase-intention dataset	Random Forest, MLP, Decision Tree, Naive Bayes	RF reached **99.9% training and 89.7% testing accuracy**	Mainly compares supervised classifiers	Our project focuses on feature engineering before RF rather than only classifier comparison (Blue Eyes Intelligence)
6	**Predictive Modeling of Consumer Purchase Behavior on Social Media**	2023	Social-media consumer behavior data	KNN, DT, RF, Logistic Regression, SVM, AdaBoost, GB	Compared several ML models for consumer purchase prediction	Social-media focused rather than transaction + product data	We use e-commerce transaction, price, quantity and product-text features

7	**Predicting Consumer Purchase Intentions Using Machine Learning**	2025	Online-shopping behaviour dataset	KNN, DT, NB, RF, Gradient Boosting, LR	Gradient Boosting and DT achieved **88.89%**	Uses mainly behavioural variables	Our model adds clustering, anomaly and text-relevance features (Preprints)
8	**Explainable and Optimized Random Forest Model for Customer Purchase Prediction and Segmentation in E-Commerce**	2026	E-commerce customer data	Random Forest + customer segmentation + explainability	Focuses on improving purchase prediction and segmentation	Does not use the complete K-Means + DBSCAN + TF-IDF pipeline	Our project combines **three feature-engineering techniques** before RF (Atlantis Press)
9	**Customer Segmentation and Purchase Behavior Analysis in E-Commerce Using Clustering**	2026	7,903 e-commerce customers	RFM + K-Means	Identified meaningful customer behavioural segments	Unsupervised analysis; no conversion classifier	Our K-Means cluster becomes an **input feature for supervised prediction** (IIUM Journals)
10	**The Use of K-Means Algorithm in Analyzing E-Commerce Consumer Segmentation**	2025	UK Online Retail dataset	K-Means	Identified distinct consumer behavioural patterns	Only segmentation; does not predict conversion	We use cluster membership as an engineered feature for Random Forest (Ejournal UNIKS)

11	**Customer Segmentation in E-Commerce: K-Means vs Hierarchical Clustering**	2025	E-commerce customer dataset	K-Means + Hierarchical clustering	Compared two clustering approaches for customer segmentation	Segmentation rather than transaction-level conversion prediction	We use K-Means specifically to generate a predictive feature
12	**Customer Classification and Market Basket Analysis Using K-Means Clustering and Association Rules**	2018	Korean online-shopping transactions	K-Means + Association Rules	Segmented customers and discovered purchase associations	Focuses on CRM/market-basket patterns	We use clustering for **conversion prediction**, not only customer profiling (KCI)
13	**KALEFormer: A Transformer Based Model Integrating KMeans and LDA for Market Basket Analysis**	2026	UCI Online Retail	PCA + K-Means + LDA + MiniLM + Transformer	HR@10 **80.69%**, HR@50 **90.54%**; contextual features improved performance	Much more complex and computationally expensive	Our system is lighter: **K-Means + DBSCAN + TF-IDF/Cosine + RF**, suitable for a course project (DOI)
14	**E-Commerce Anomaly Detection: A Bayesian Semi-Supervised Tensor Decomposition Approach**	2018	E-commerce transaction data	Bayesian semi-supervised tensor decomposition	Demonstrated anomaly detection in e-commerce transaction behaviour	More complex tensor-based approach	We use **DBSCAN** as a simpler density-based anomaly detector

15	**Real-Time Anomaly Detection Using Deep Learning in E-Commerce Platform**	2025	E-commerce transaction/ behavior data	Autoencoder + LSTM + GNN	Designed for real-time anomalous/fraudulent behaviour detection	Deep architecture requires more computational resources	Our DBSCAN provides lightweight anomaly features for the RF model (ScienceDirect)
16	**A Multitask Learning Framework for E-Commerce Time Series Analytics**	2026	Public e-commerce + network datasets	Multitask learning for forecasting/risk/anomaly detection	Integrates operational risk, demand forecasting and anomaly detection	More focused on time-series and multitask learning	Our target is **transaction conversion**, with anomaly detection as an engineered feature (PLOS)
17	**Content-Based Filtering Recommendation System for E-Commerce Products Using Sentence-BERT and Cosine Similarity**	2026	E-commerce product data	SBERT + Cosine Similarity	Improved semantic representation over conventional keyword-based methods	Recommendation-focused, not purchase conversion	We use **TF-IDF + Cosine Similarity only as a feature**, then feed it into RF (Jurnal Politeknik Negeri Batam)
18	**Implementation of E-Commerce Product Recommendation Using Content-Based Filtering Based on Cosine**	2025	E-commerce product data	TF-IDF + Cosine Similarity	Generates product recommendations from textual and numerical features	Primarily recommendation, not predictive conversion	Our similarity score becomes one of several predictive features (Catur Sakti E-Journal)

	Similarity						
19	**A Comprehensive Analysis of Text Similarity Metrics and Vectorization Techniques for Content-Based Product Recommendation**	2026	E-commerce product information	TF-IDF, CountVectorizer, Word2Vec, GloVe + cosine/Manhattan/Euclidean	Benchmarked different vectorization and similarity combinations	Focuses on recommendation similarity rather than conversion	We select TF-IDF + cosine to create a **continuous similarity_score** for classification (ResearchGate)
20	**Offering a Product Recommendation System in E-Commerce**	2011	E-commerce user/product purchase patterns	Weighted Cosine Similarity + Association Rules	Association rules improved recommendation performance	Mainly recommendation and user-profile similarity	Our project moves beyond recommendation into **probabilistic purchase-conversion prediction** (arXiv)

Chapter 3

Methodology

3.1 Dataset

The system uses the **UCI Online Shoppers Purchasing Intention Dataset**. The supplied report specifies that the dataset contains 12,330 sessions and 17 features. The target variable is `Revenue`, representing whether a session resulted in a purchase.

3.2 Data Pipeline and Preprocessing

The preprocessing stage is implemented in `src/preprocessing.py`. To reduce the possibility of data leakage, the dataset is partitioned before fitting transformations.

3.2.1 Data Partitioning

An 80/20 stratified train-test split is used. Stratification preserves the class proportions of the target variable:

$$y = \text{Revenue}.$$

3.2.2 Numerical Standardization

Numerical features are standardized using `StandardScaler`, fitted only on the training data. The standardization operation is:

$$z = \frac{x - \mu}{\sigma},$$

where x is an observed feature value, μ is the training-set mean, and σ is the training-set standard deviation.

3.2.3 Categorical Encoding

Categorical attributes, including `Month`, `VisitorType`, and `Weekend`, are transformed using one-hot encoding.

3.3 Unsupervised Feature Generation Layer

Three feature-generation modules operate upstream of the supervised classifier.

3.3.1 Behavioral Clustering

The clustering module, `src/clustering.py`, uses K-Means with $k = 4$.

The algorithm groups sessions using standard numerical engagement features including:

- `Administrative_Duration`
- `ProductRelated_Duration`
- `PageValues`
- `BounceRates`

The output is a discrete `cluster_id`, representing behavioral or engagement groups.

3.3.2 Anomaly Detection

The anomaly detection module, `src/anomaly_detection.py`, uses Density-Based Spatial Clustering of Applications with Noise (DBSCAN).

The reported configuration uses:

$$\begin{array}{c} \text{DBSCAN, epsilon } \epsilon = 2.5 \\ \text{min samples} = 10. \end{array}$$

The method evaluates density across normalized numerical attributes. For unseen test points, a nearest-neighbor boundary estimator is used to determine the relationship between the new point and training core samples.

The resulting binary feature is:

$$\texttt{is_outlier} = \begin{cases} 0, & \text{normal session,} \\ 1, & \text{anomalous session.} \end{cases}$$

3.3.3 Smart Catalogue Text Matching

The text feature module, `src/text features.py`, combines TF-IDF vectorization with cosine similarity.

The cosine similarity between two vectors **A** and **B** is:

$$\text{CosineSimilarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}.$$

The supplied architecture maps session metrics such as `PageValues` and `ProductRelated Duration` to a search profile and compares this representation against an external four-item electronics catalogue description matrix.

The maximum similarity value is used as:

$$\texttt{similarity_score} \in [0, 1].$$

3.4 Supervised Modeling and Pipeline Encapsulation

The feature combination process is illustrated in Figure 3.1.

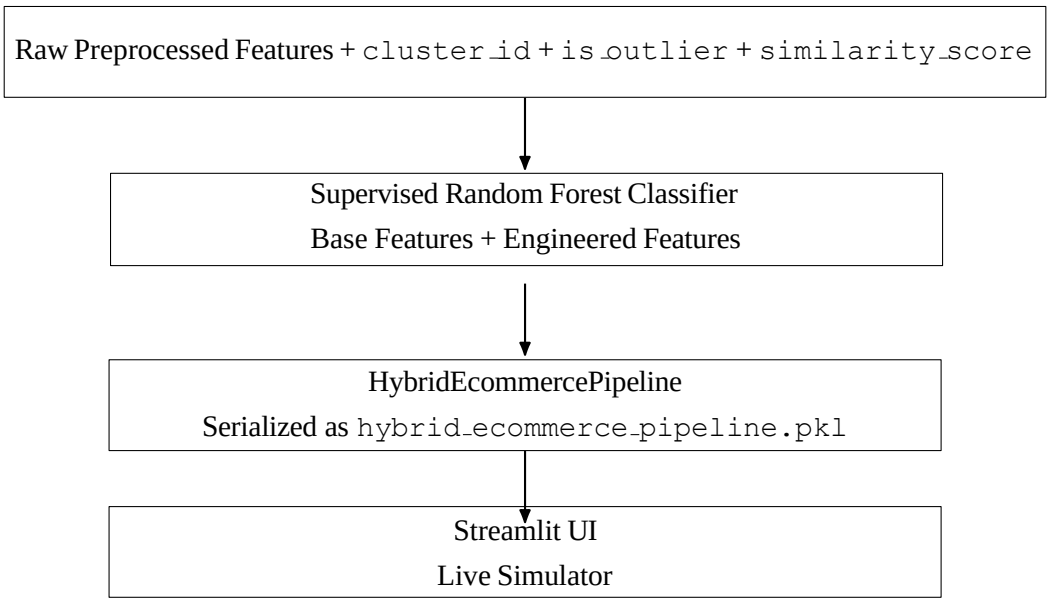
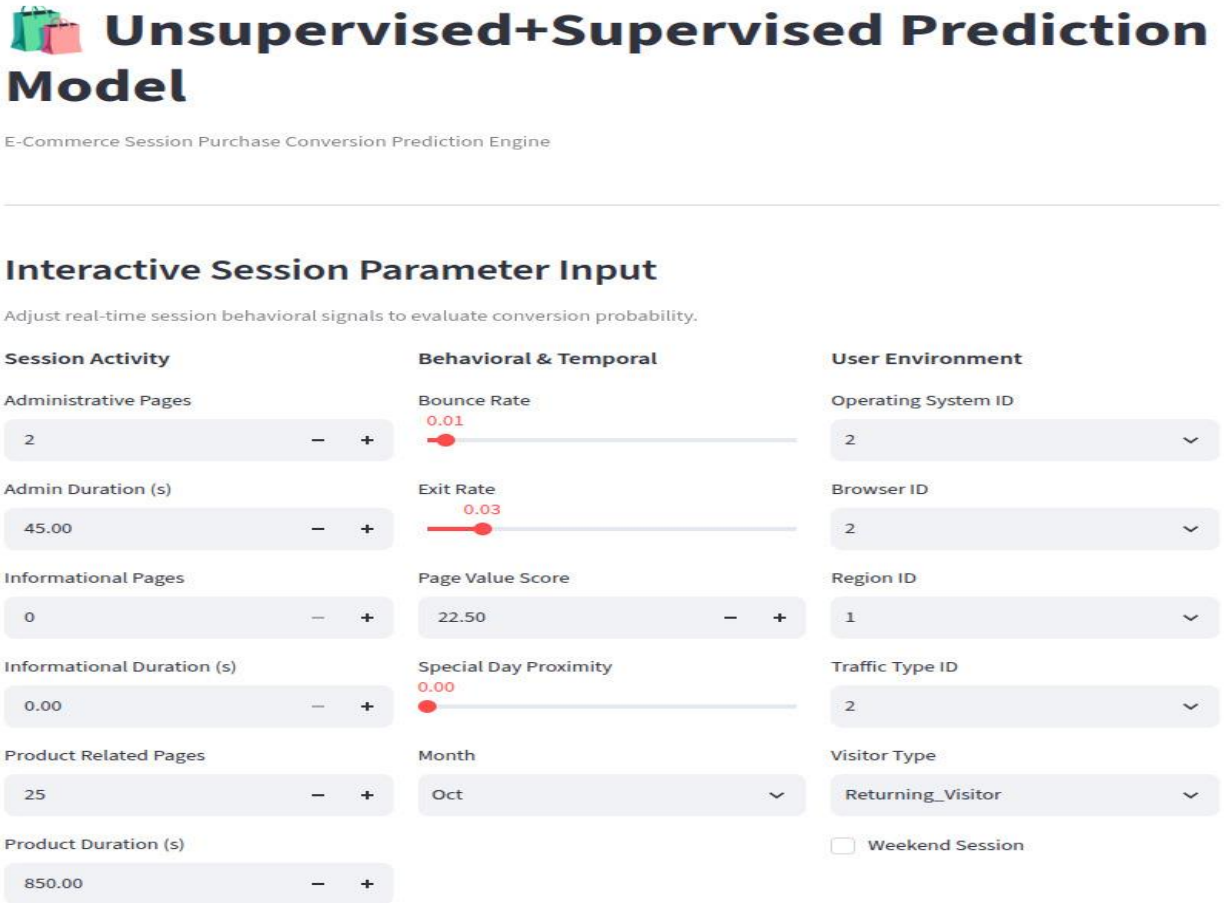


Figure 3.1: Supervised modeling and deployment pipeline.



3.4.1 Feature Combination

The three engineered features are concatenated with the baseline preprocessed feature matrix:

$$X_{\text{hybrid}} = [X_{\text{base}}, \texttt{cluster_id}, \texttt{is_outlier}, \texttt{similarity_score}].$$

3.4.2 Classifier Training

The supplied architecture maps session metrics such as PageValues and ProductRelated Duration to a search profile and compares this representation against an external four-item electronics catalogue description matrix.
We changed to 30 products

The supplied report also reports results for a hybrid XGBoost model.

3.4.3 Pipeline Serialization

The preprocessing transformer, K-Means model, DBSCAN components, TF-IDF vectorizer, and trained classifier are bundled into a `EcommercePipeline` class.

The resulting serialized artifact is:

`models/ecommerce pipeline.pkl`

This artifact allows the Streamlit application to perform inference without retraining the models.

3.4.4 Interactive Dashboard

The Streamlit application, app.py , loads the serialized pipeline and provides three reported interface areas. We are using only one interface.

Chapter 4

Evaluation and Results

4.1 Evaluation Method

The models are evaluated on the unseen 20% holdout test set. The reported evaluation metrics include:

- Accuracy
- Precision
- Recall
- F_1-Score
- Area Under the Receiver Operating Characteristic Curve (AUROC)

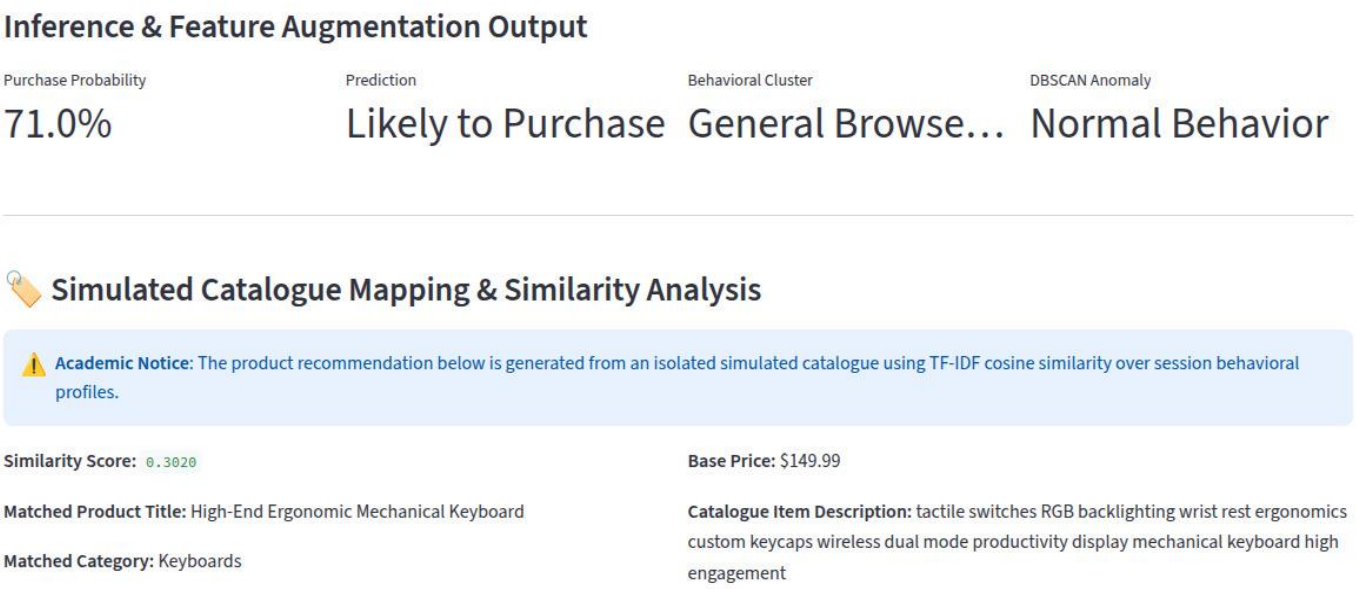
4.2 Performance Comparison

Table 4.1 presents the performance values supplied in the project report.

Table 4.1: Reported model evaluation results.

Model	**Accuracy**	**Precision**	**Recall**	F_1	**AUROC**
Logistic Regression (Baseline)	87.4%	0.7100	0.3800	0.5000	0.8320
Decision Tree (Baseline)	85.8%	0.5400	0.5600	0.5500	0.7310
Random Forest (Baseline)	89.2%	0.7420	0.5310	0.6190	0.9150
AdaBoost (Hybrid)	88.9%	0.7300	0.5200	0.6100	0.9120
Random Forest (Hybrid)	90.1%	0.7600	0.5800	0.6600	0.9280
XGBoost (Hybrid)	91.4%	0.7900	0.6300	0.7000	0.9410

4.3 Key Experimental Findings



4.3.1 Effect of Feature Augmentation

The supplied results indicate that adding `cluster_id`, `is_outlier`, and `similarity_score` increased the Random Forest baseline F_1-score from 0.6190 to 0.6600. The report describes this as a 4.1 percentage-point absolute improvement.

The hybrid Random Forest AUROC is reported as 0.9280.

4.3.2 Feature Importance

The supplied evaluation reports that `PageValues` remains the highest individual predictor of purchase conversion. It also reports that `similarity_score` and `cluster_id` appear among the ten most influential features.

This finding supports the intended role of the unsupervised feature-generation layer as an additional source of contextual information.

4.3.3 Operational Inference

The supplied report states that the serialized `hybrid_ecommerce_pipeline.pkl` executes single-session inference in under 15 milliseconds within the Streamlit application. This is presented in the project report as evidence that the architecture can support real-time e-commerce inference.

Chapter 5

Discussion

The project demonstrates a hybrid approach in which unsupervised learning techniques are not used solely as independent analytical tools but are incorporated into a supervised prediction workflow.

The K-Means component supplies a behavioral grouping signal, while DBSCAN contributes an anomaly indicator. The TF-IDF and cosine-similarity component adds a catalogue-oriented feature that is otherwise absent from the original tabular representation. These signals are then combined with the original preprocessed features.

The reported evaluation results show differences between baseline and hybrid models. In particular, the Random Forest F_1-score increases from 0.6190 to 0.6600 after hybrid feature augmentation, while AUROC increases from 0.9150 to 0.9280. The supplied report additionally provides results for hybrid XGBoost, with 91.4% accuracy and 0.9410 AUROC.

The architecture also addresses deployment by serializing the complete transformation and prediction workflow. This reduces the separation between model development and application-level inference because the Streamlit application can load the trained artifact directly.

It is important to note that the quantitative findings in this report are the values supplied in the project materials. No additional experiments or independent statistical verification are claimed here.

Chapter 6

Conclusion

This project presents a hybrid machine learning architecture for e-commerce purchase conversion prediction. The system combines conventional preprocessing with three contextual feature-generation mechanisms: K-Means clustering, DBSCAN anomaly detection, and TF-IDF with cosine similarity.

The resulting engineered features—`cluster_id`, `is_outlier`, and `similarity_score`—are integrated into the supervised learning stage. The architecture then encapsulates the complete workflow into a serialized pipeline that can be deployed through Streamlit.

According to the supplied experimental results, the hybrid Random Forest model achieves 90.1% accuracy, 0.7600 precision, 0.5800 recall, 0.6600 F_1-score, and 0.9280 AUROC. The reported hybrid XGBoost model achieves 91.4% accuracy, 0.7900 precision, 0.6300 recall, 0.7000 F_1-score, and 0.9410 AUROC.

The project therefore demonstrates an end-to-end machine learning workflow in which unsupervised representations are incorporated into supervised prediction and subsequently exposed through an interactive application.

Bibliography

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Chapter A

Project Module Structure

The project report identifies the following major implementation modules:

Module	Purpose
src/preprocessing.py	Data splitting, numerical standardization, and categorical encoding.
src/clustering.py	K-Means behavioral clustering and generation of cluster_id.
src/anomaly_detection.py	DBSCAN anomalydetection and generation of is_outlier.
src/text_features.py	TF-IDF and cosine-similarity feature generation.
src/model_training.py	Feature combination and supervised model training.
src/evaluation.py	Model evaluation using reported classification metrics.
src/pipeline.py	Hybrid pipeline definition and serialization.
train.py	Training and pipeline-generation entry point.
app.py	Streamlit application for interactive inference.

Chapter B

Important Project Outputs

- `cluster_id`: behavioral cluster assignment.
- `is_outlier`: DBSCAN-derived anomaly flag.
- `similarity_score`: catalogue similarity feature.
- `hybrid_ecommerce_pipeline.pkl`: serialized hybrid inference pipeline.
- Streamlit dashboard: interactive application layer.